

Note: if the system is an ideal gas, then B, C and D are correct statements.

14 B Amplitude = 5.0 cm

$$a = -\omega^2 x, \quad 10 = (2\pi/T)^2 (0.050) \rightarrow T = 0.44 \text{ s}$$

15 A $\frac{1}{2} m v^2 = \frac{1}{2} k A^2 \left[\frac{1}{2} \left(\frac{A^2}{L} \right) \right]$